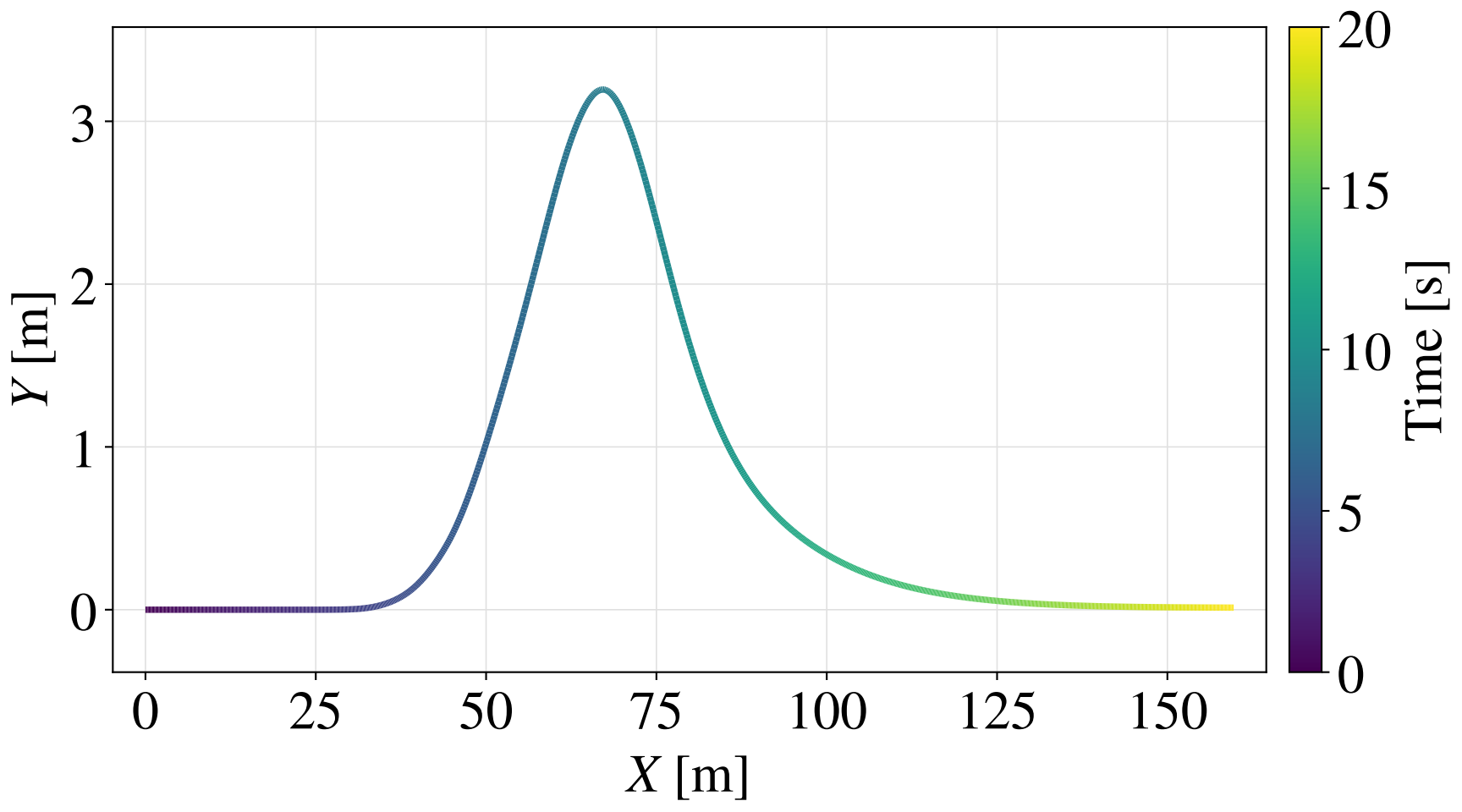
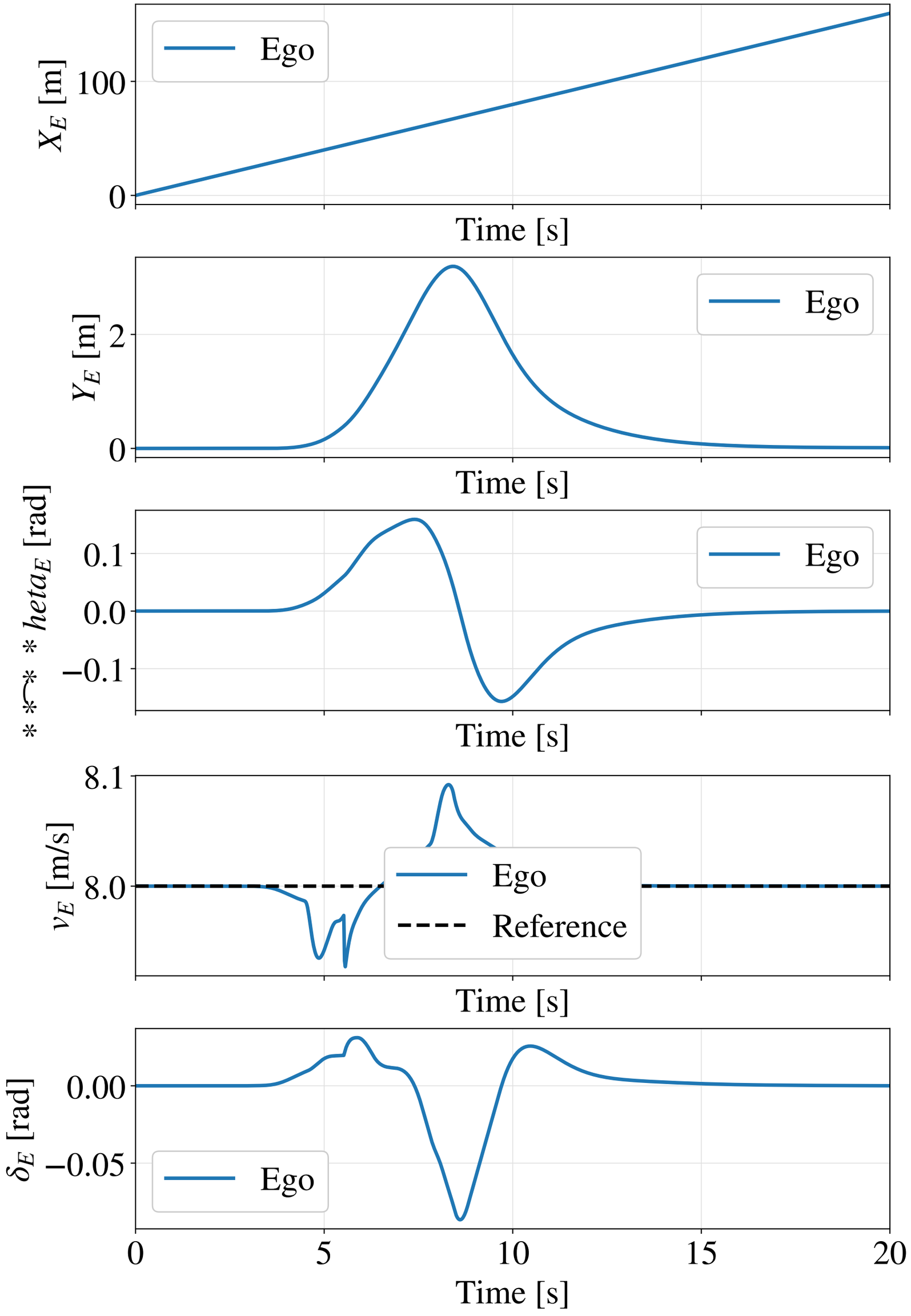
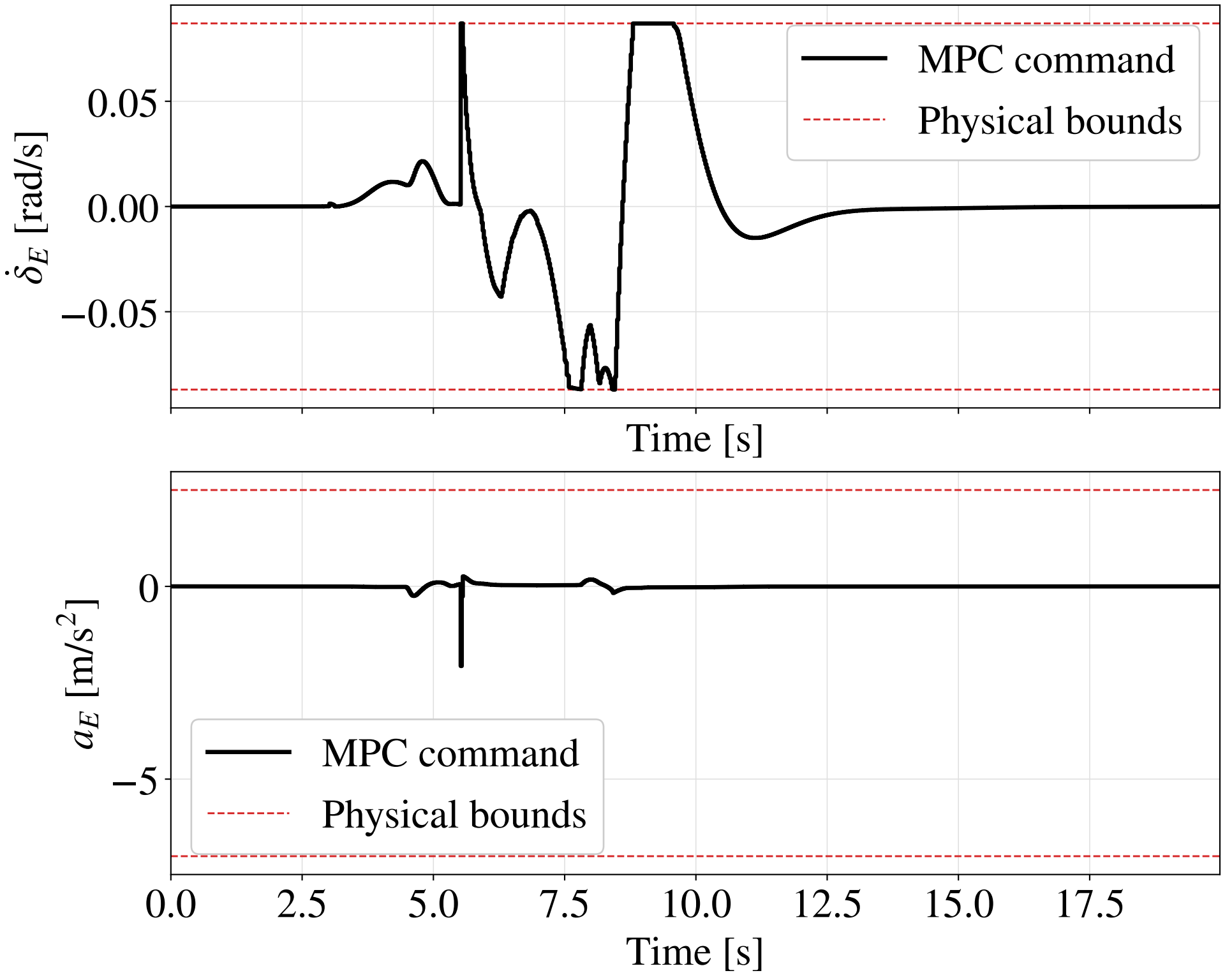
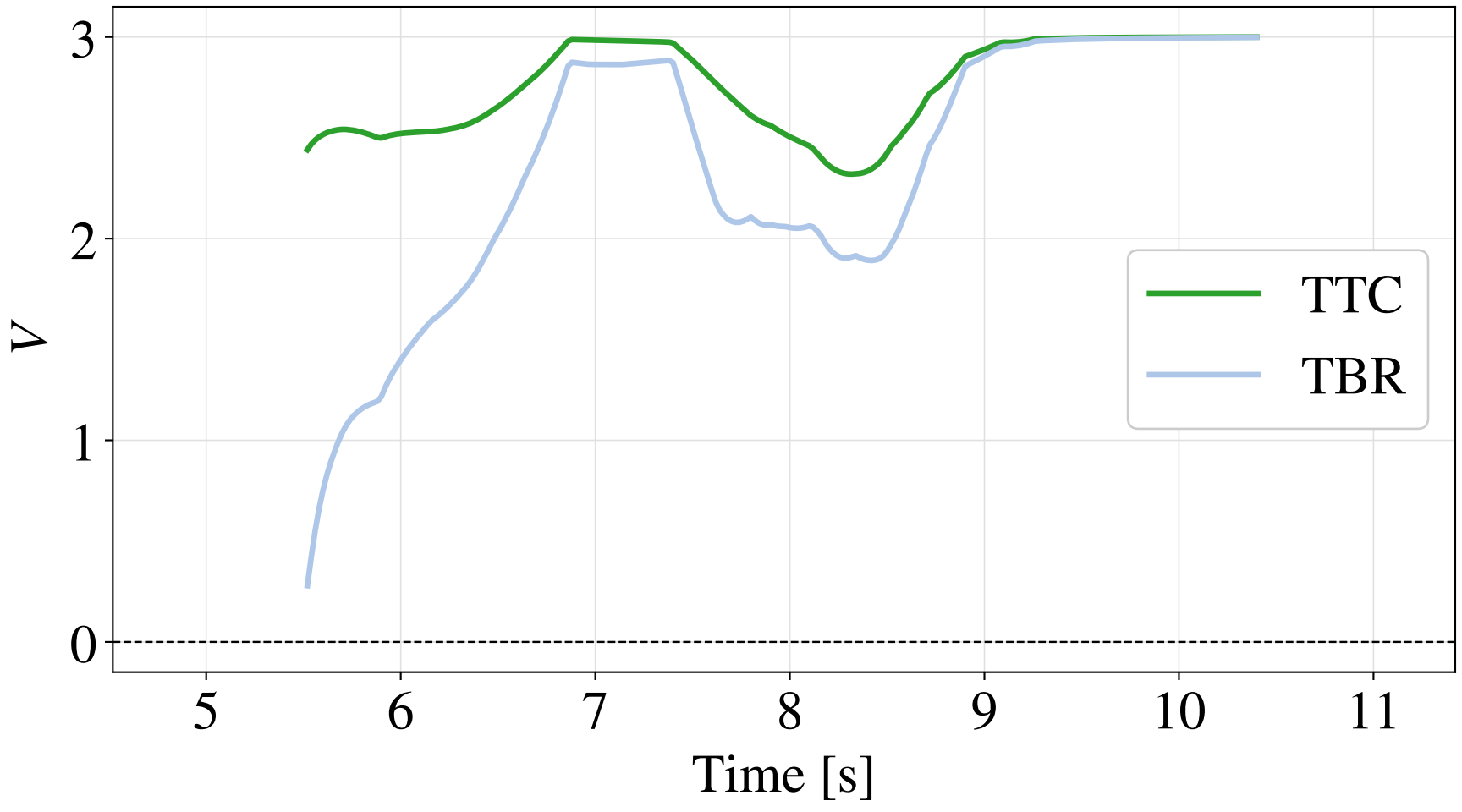
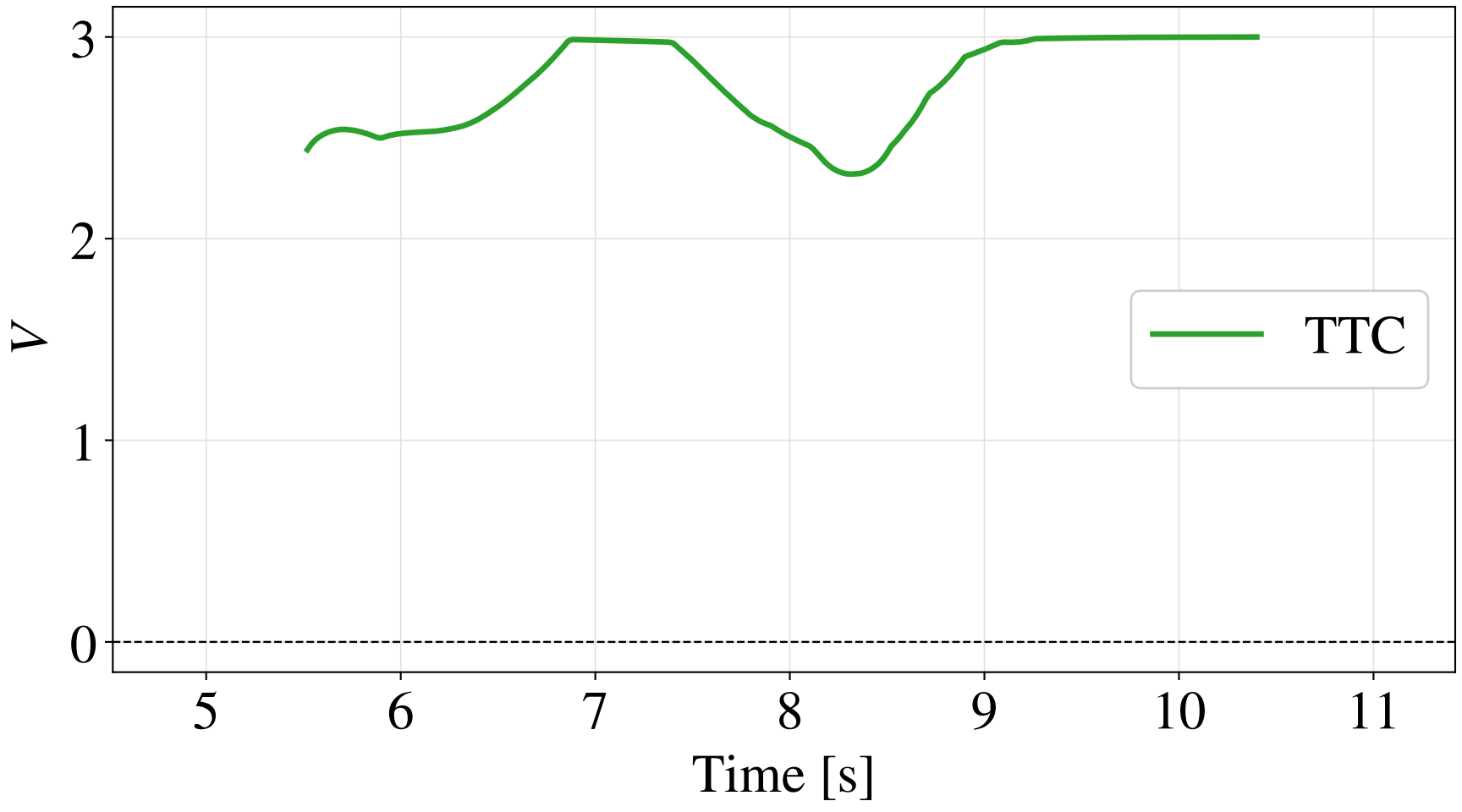
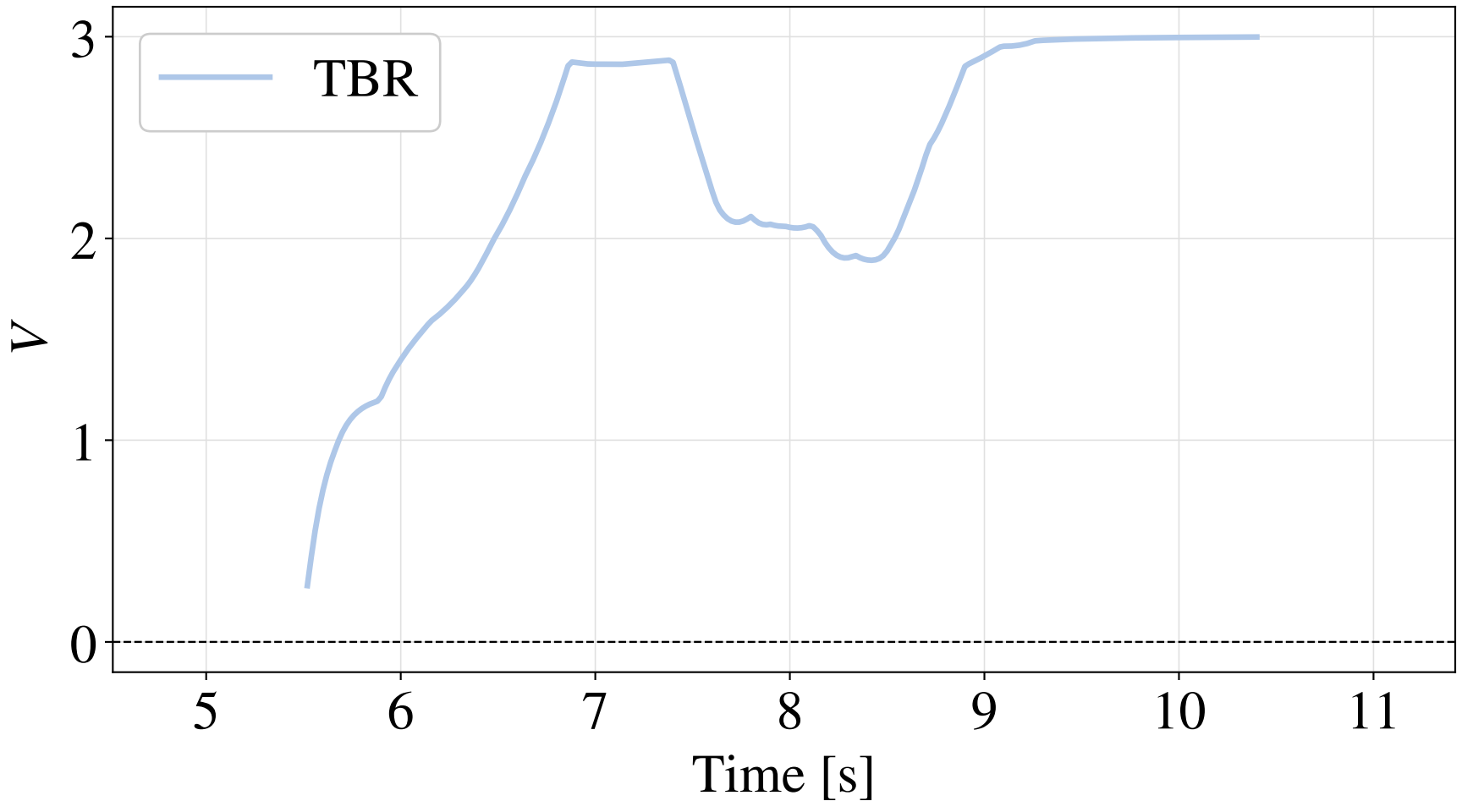

3
2
1
0
Y [m]
0
25
50
75
100
125
150
X [m]
20
15
10
5
0
Time [s]

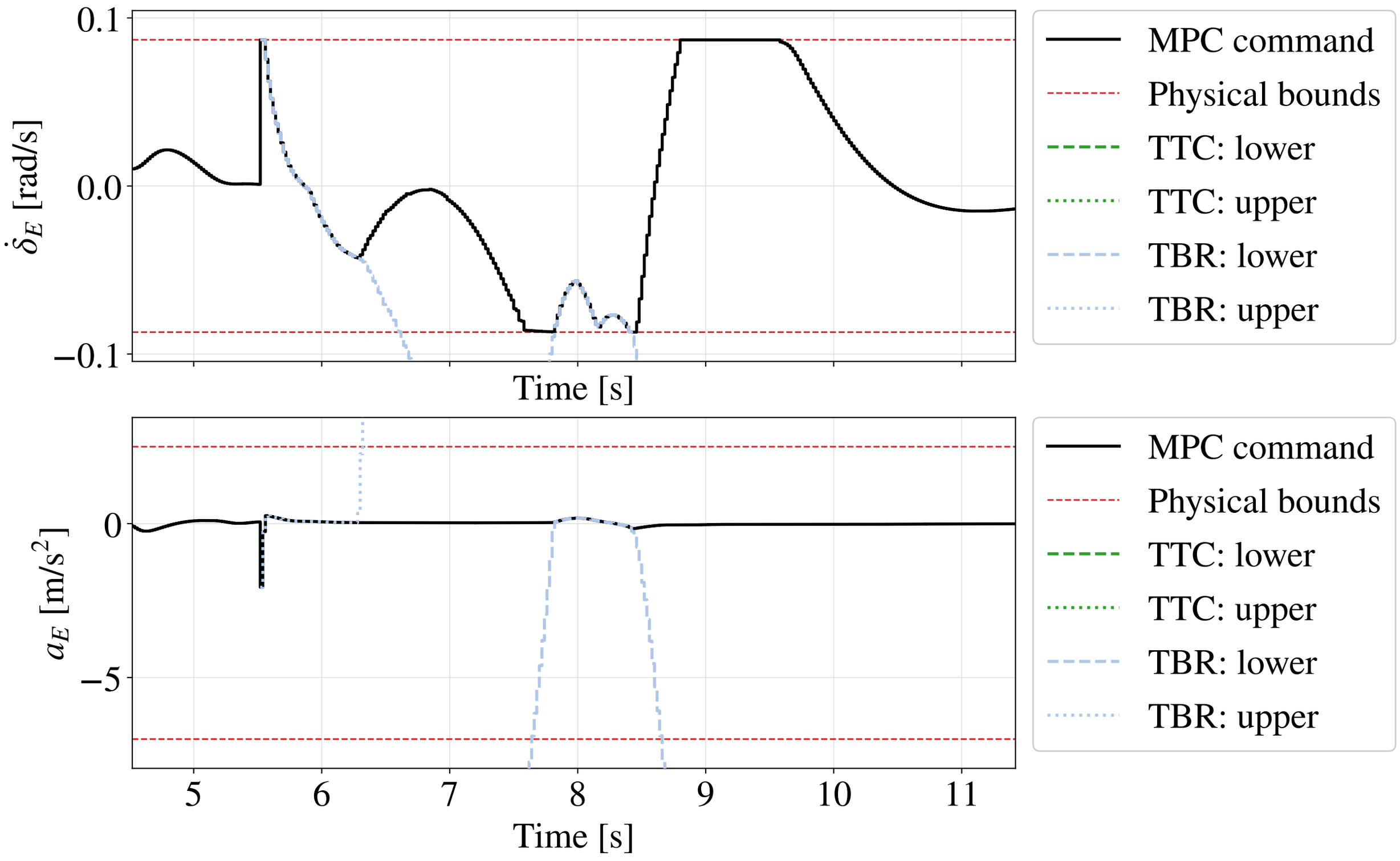

Ego
X_E [m]
100
0
Time [s]
Ego
Y_E [m]
2
0
Time [s]
Ego
ψ_E [rad]
0.1
0.0
−0.1
Time [s]
Ego
Reference
v_E [m/s]
8.1
8.0
Time [s]
Ego
δ_E [rad]
0.00
−0.05
0
5
10
15
20
Time [s]

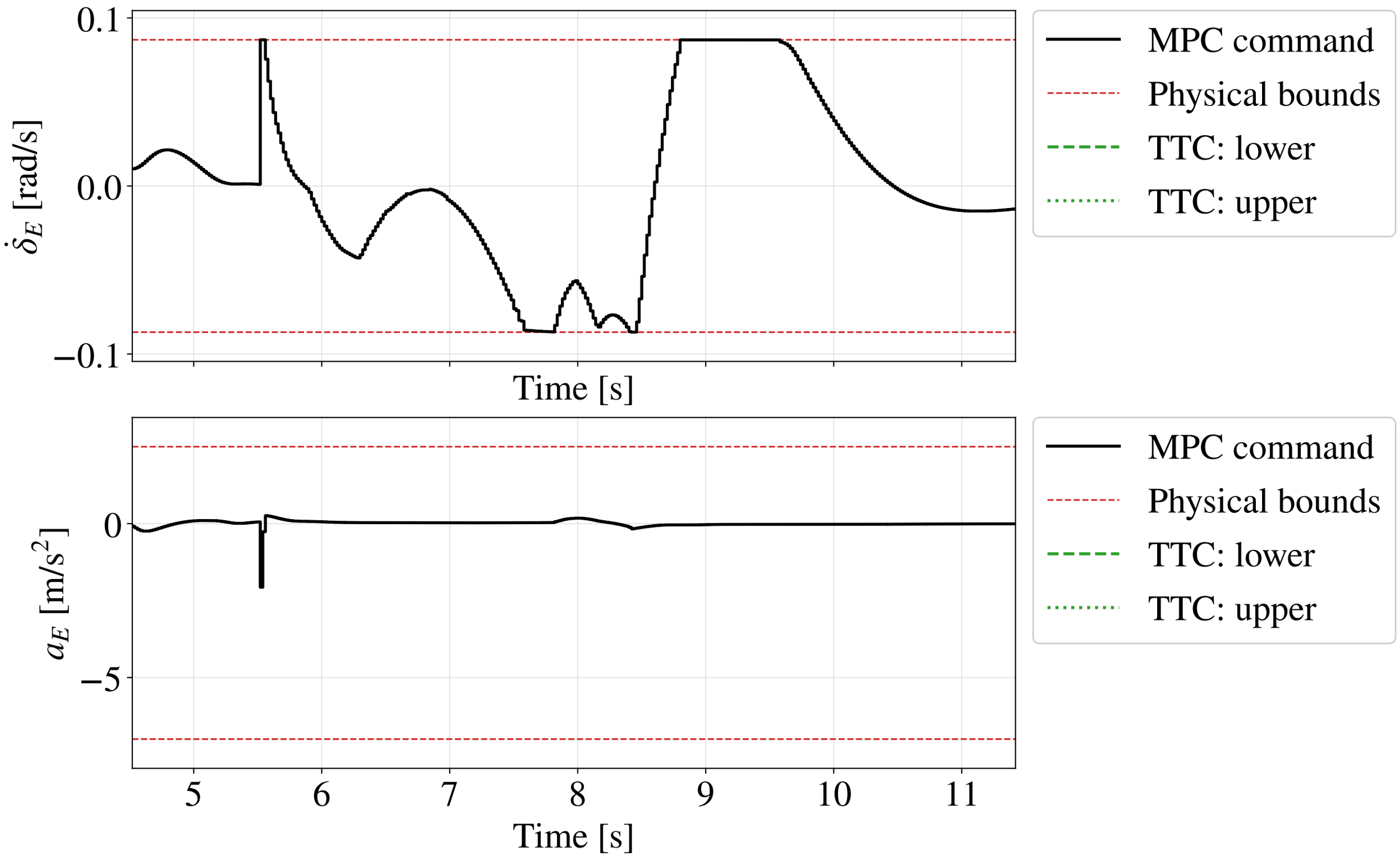

MPC command
Physical bounds
0.05
0.00
−0.05
$\dot{\delta}_E$ [rad/s]
Time [s]
0
−5
a_E [m/s^2]
MPC command
Physical bounds
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

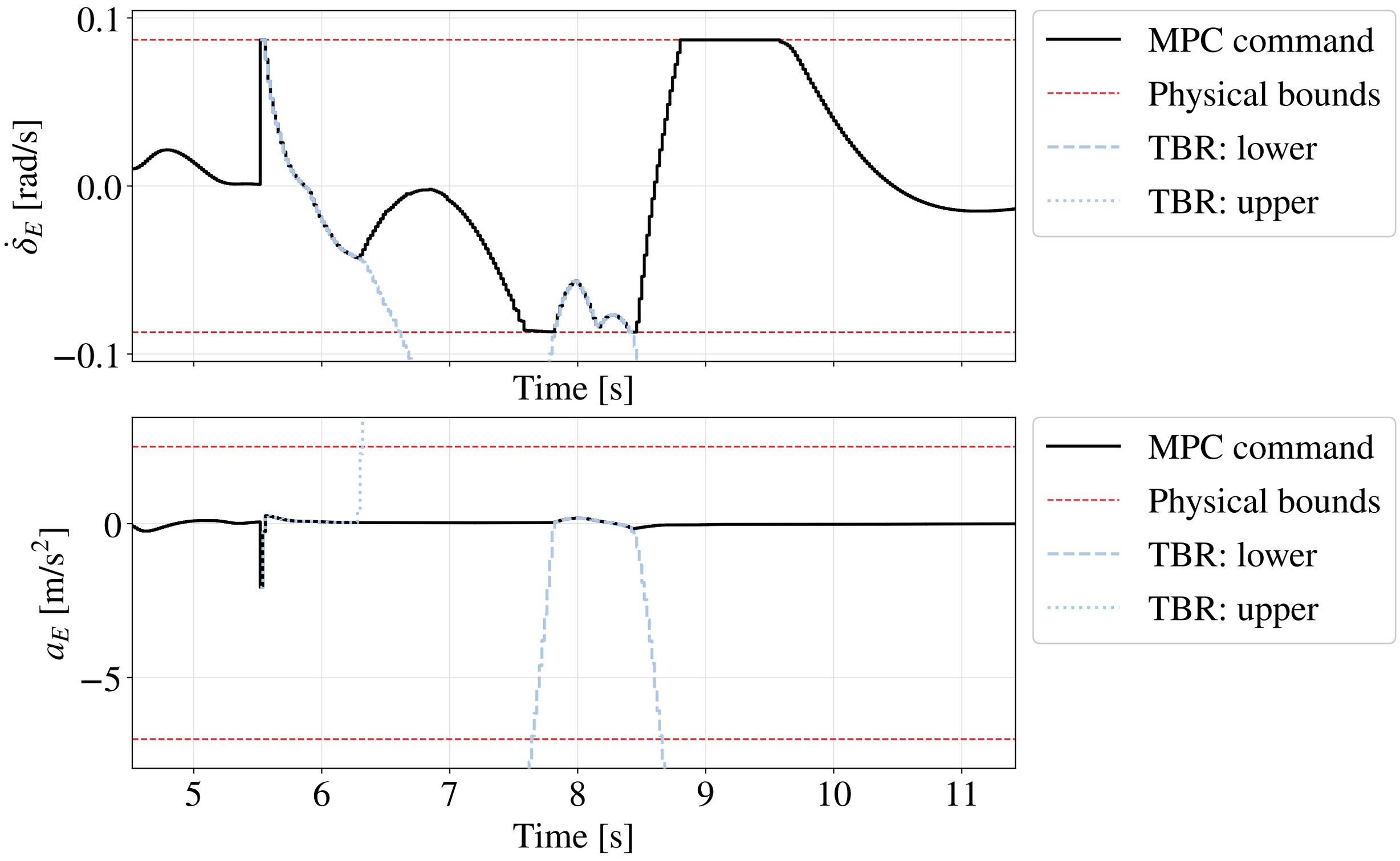

3
2
1
0
V
5
6
7
8
9
10
11
Time [s]
TTC
TBR

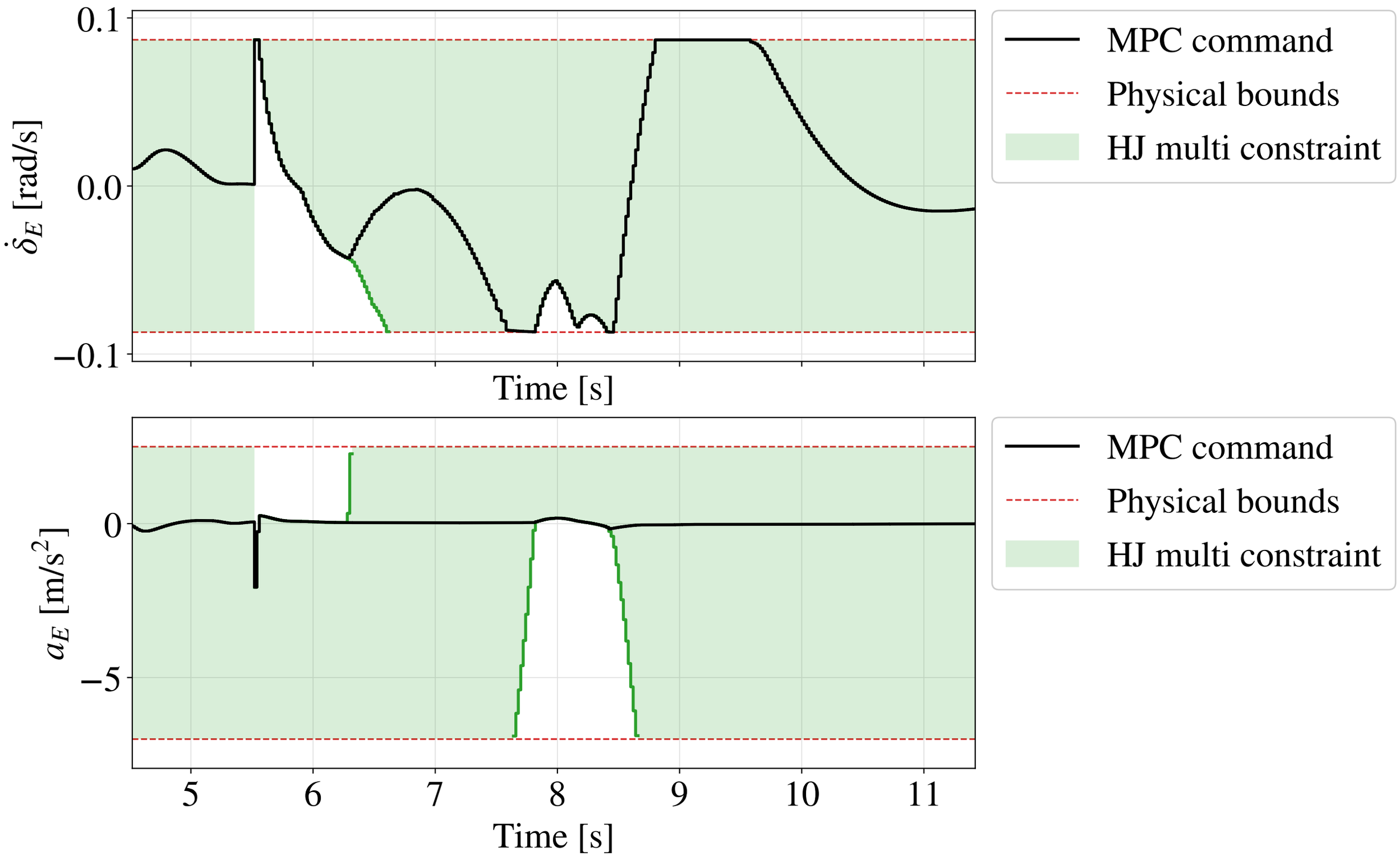

3
2
1
0
V
5
6
7
8
9
10
11
Time [s]
TTC

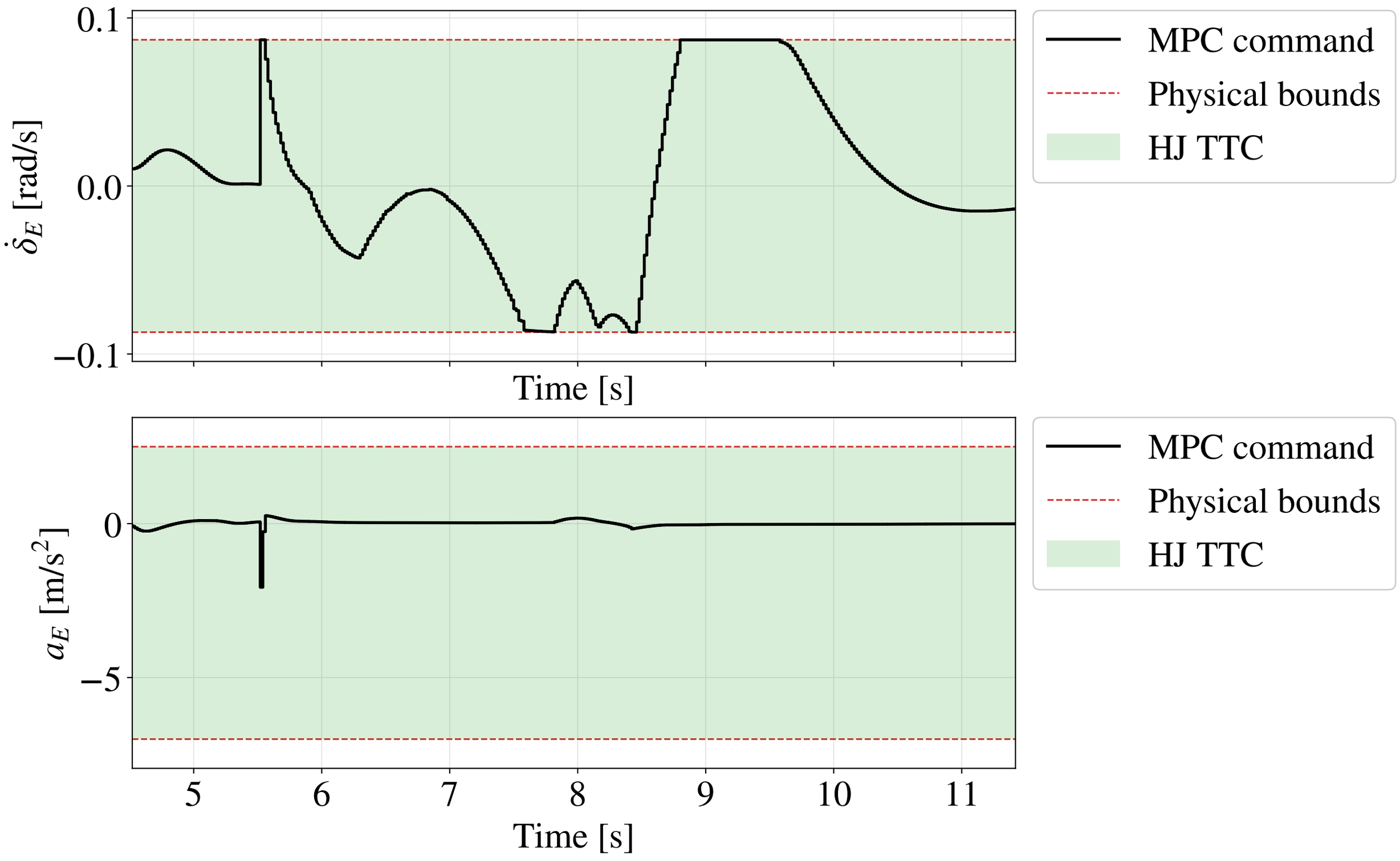

TBR
3
2
1
0
V
5
6
7
8
9
10
11
Time [s]

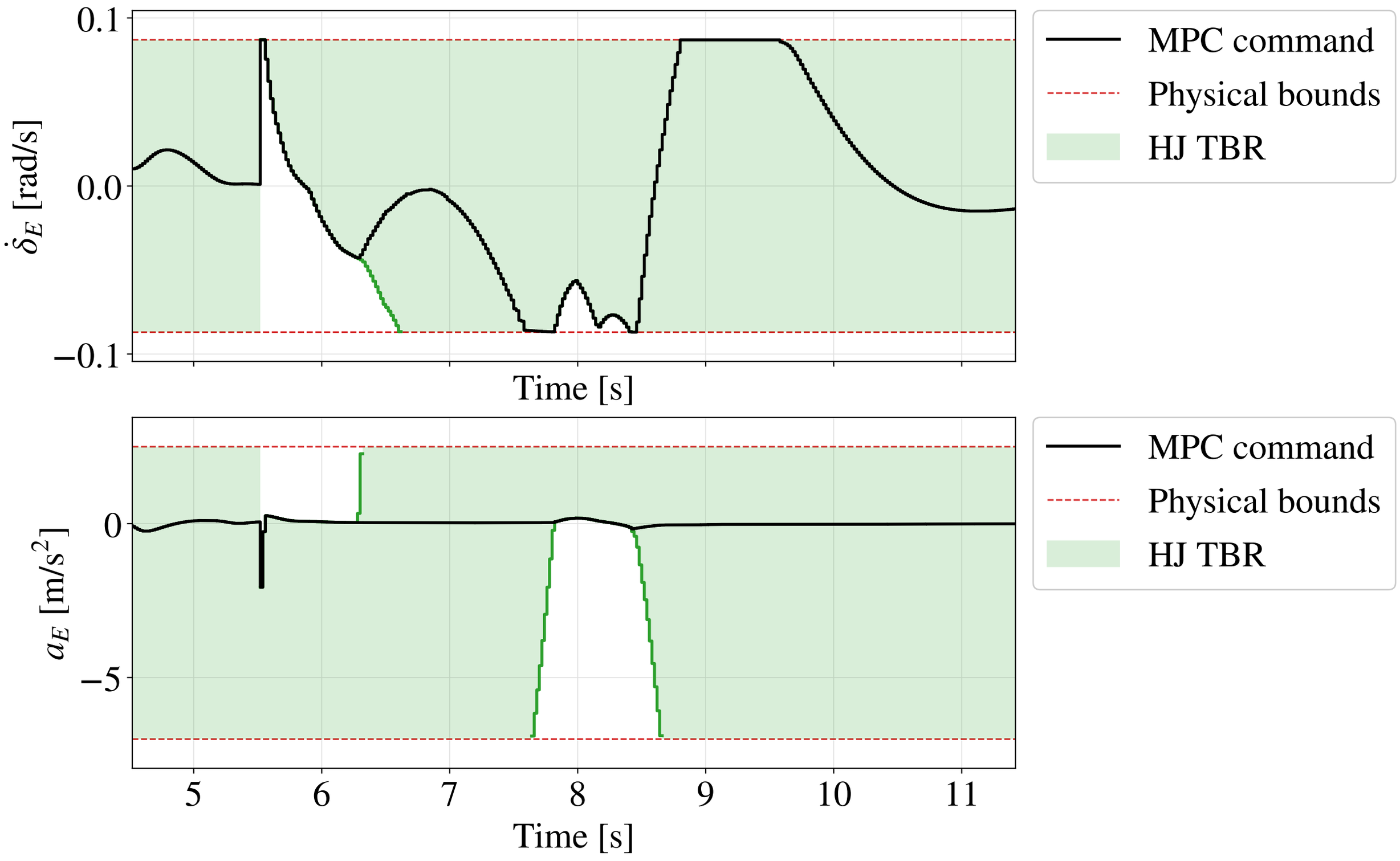

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
TTC: lower
TTC: upper
TBR: lower
TBR: upper
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
TTC: lower
TTC: upper
TBR: lower
TBR: upper

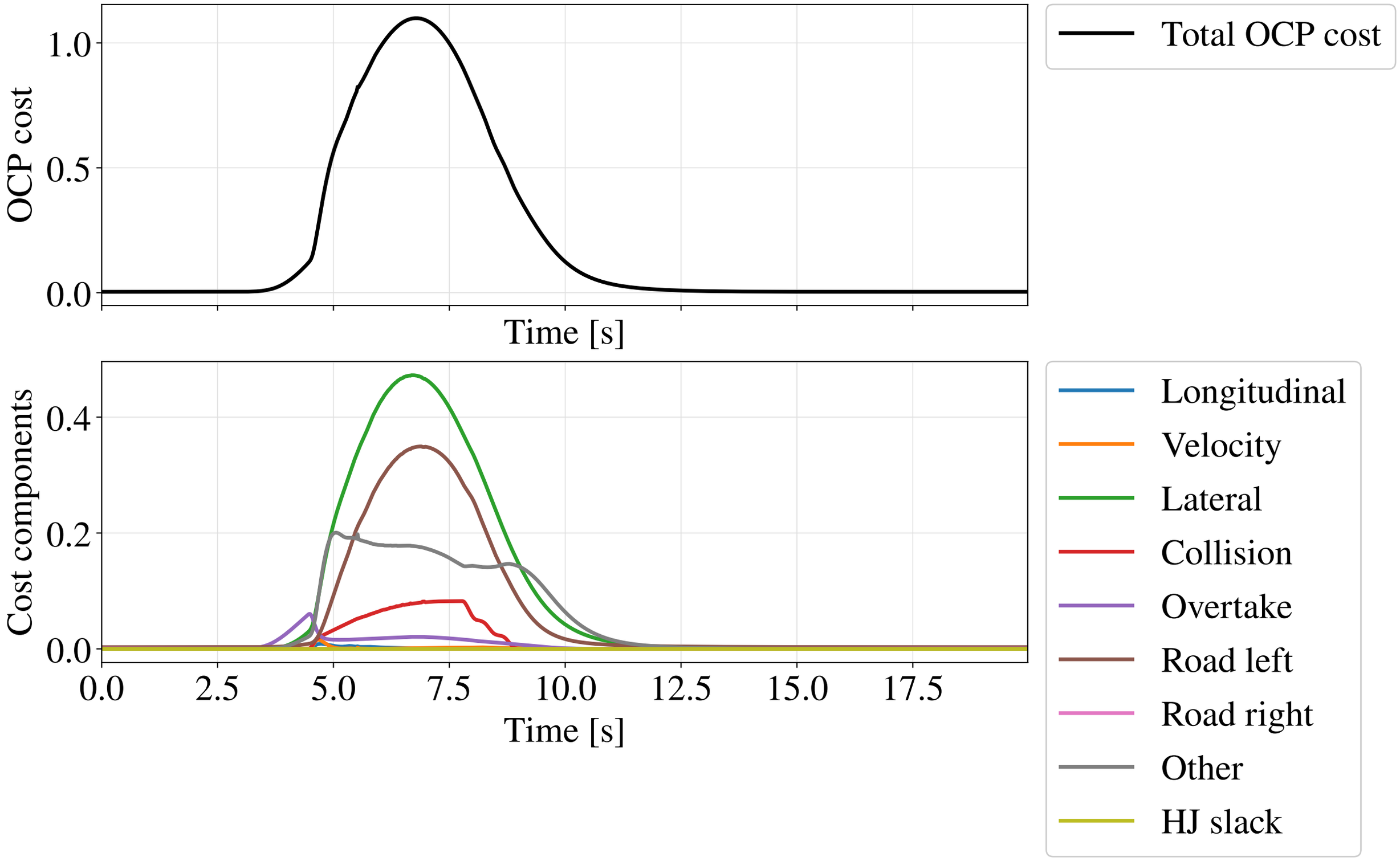

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
TTC: lower
TTC: upper
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
TTC: lower
TTC: upper

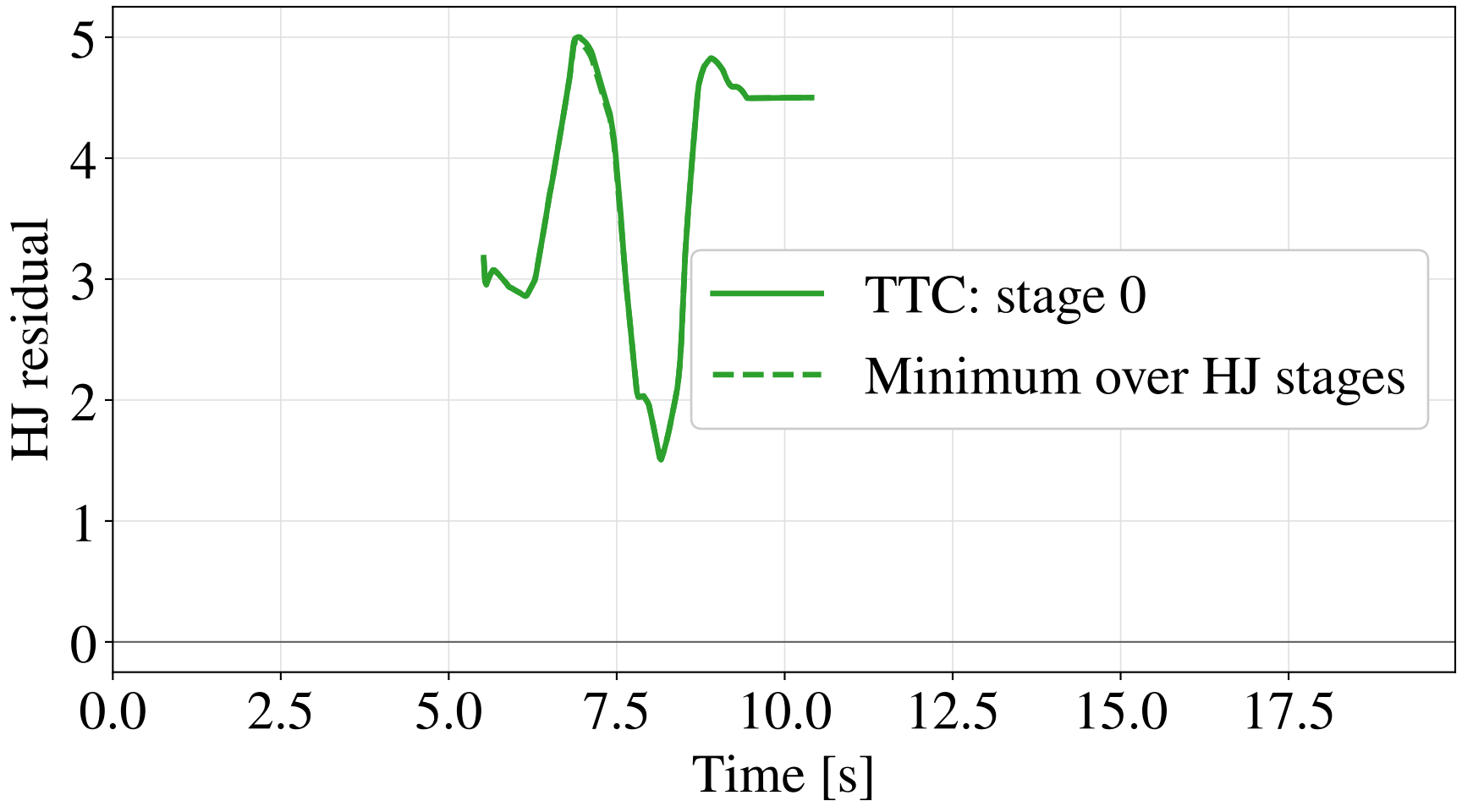

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
TBR: lower
TBR: upper
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
TBR: lower
TBR: upper

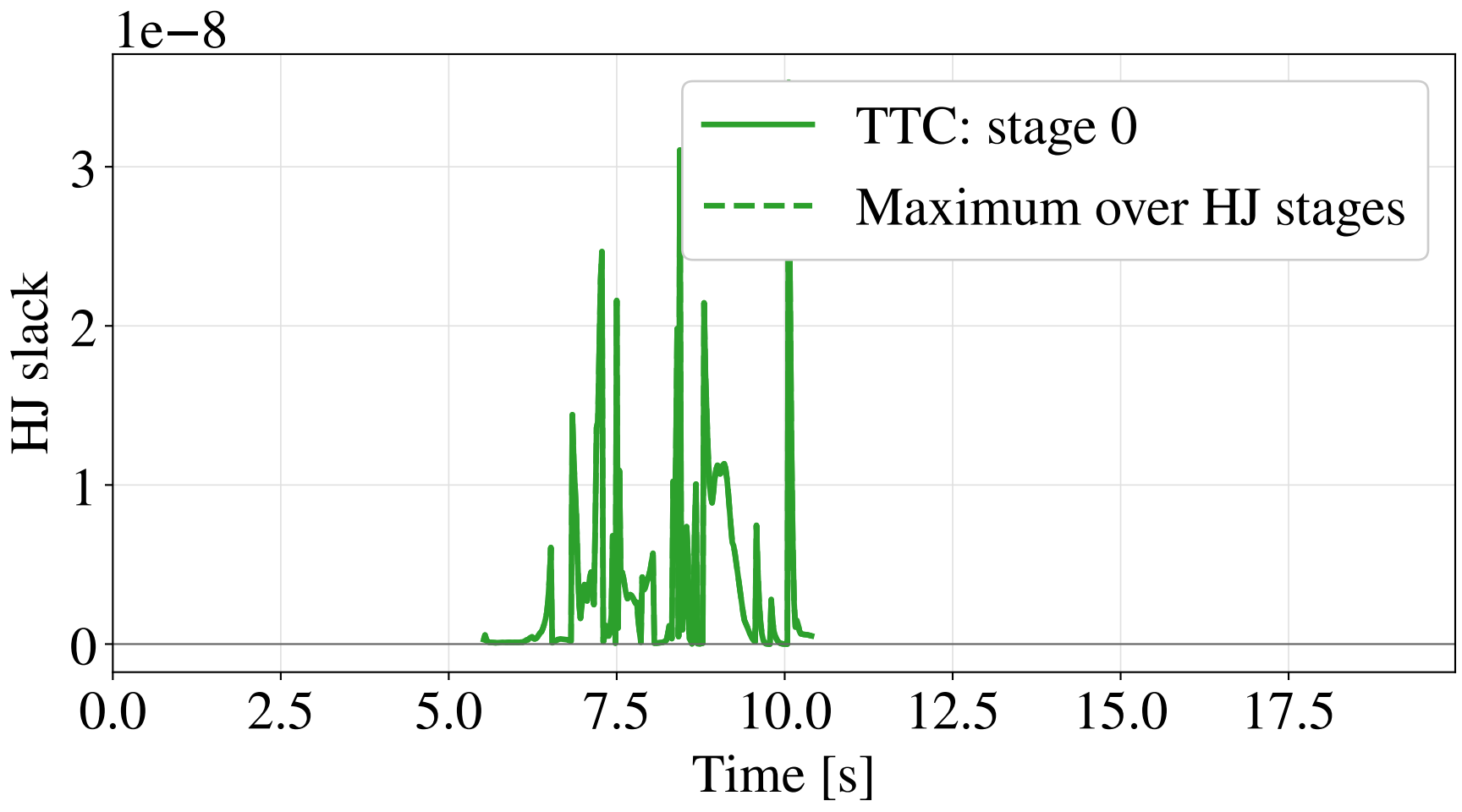

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ multi constraint
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
HJ multi constraint

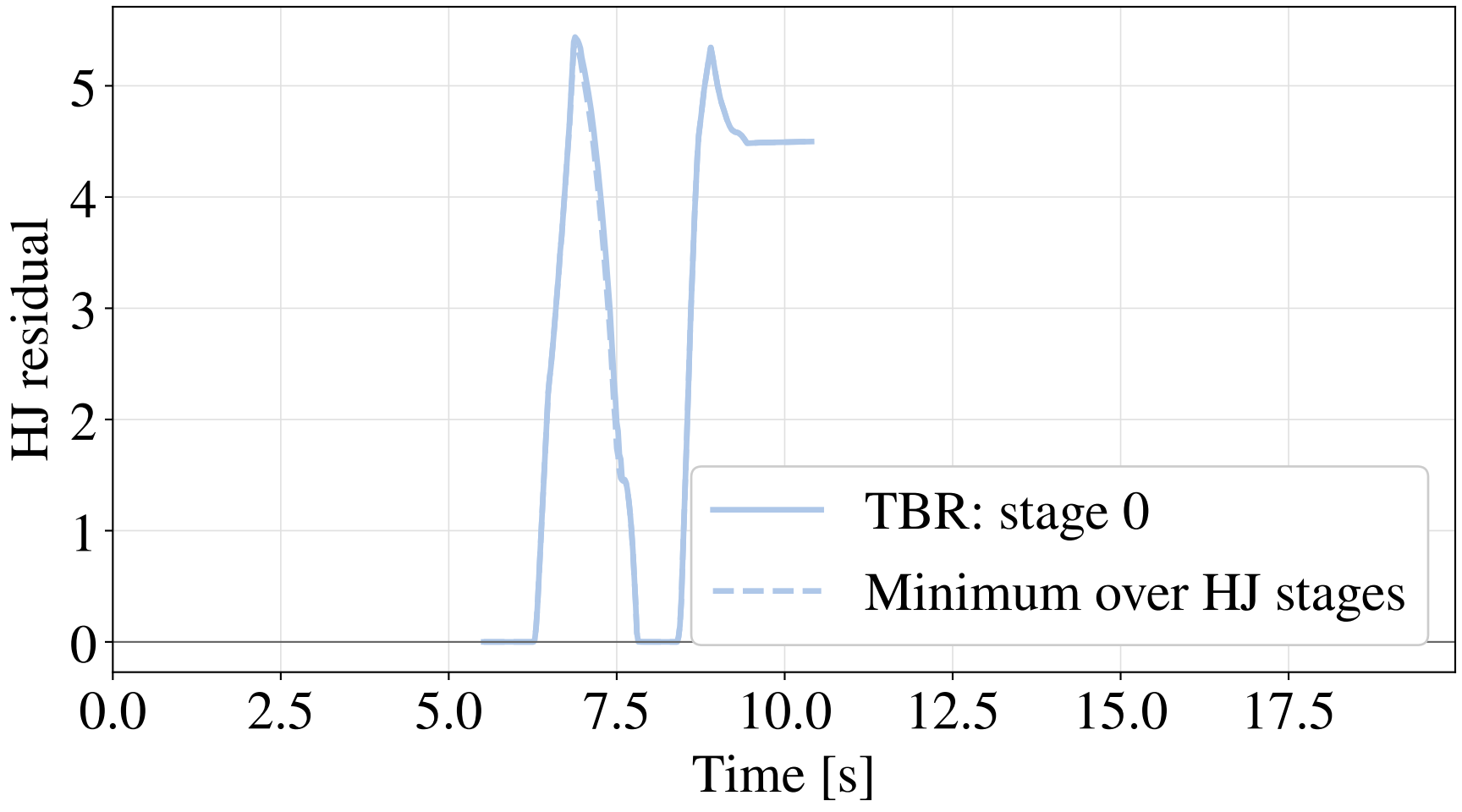

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ TTC
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
HJ TTC

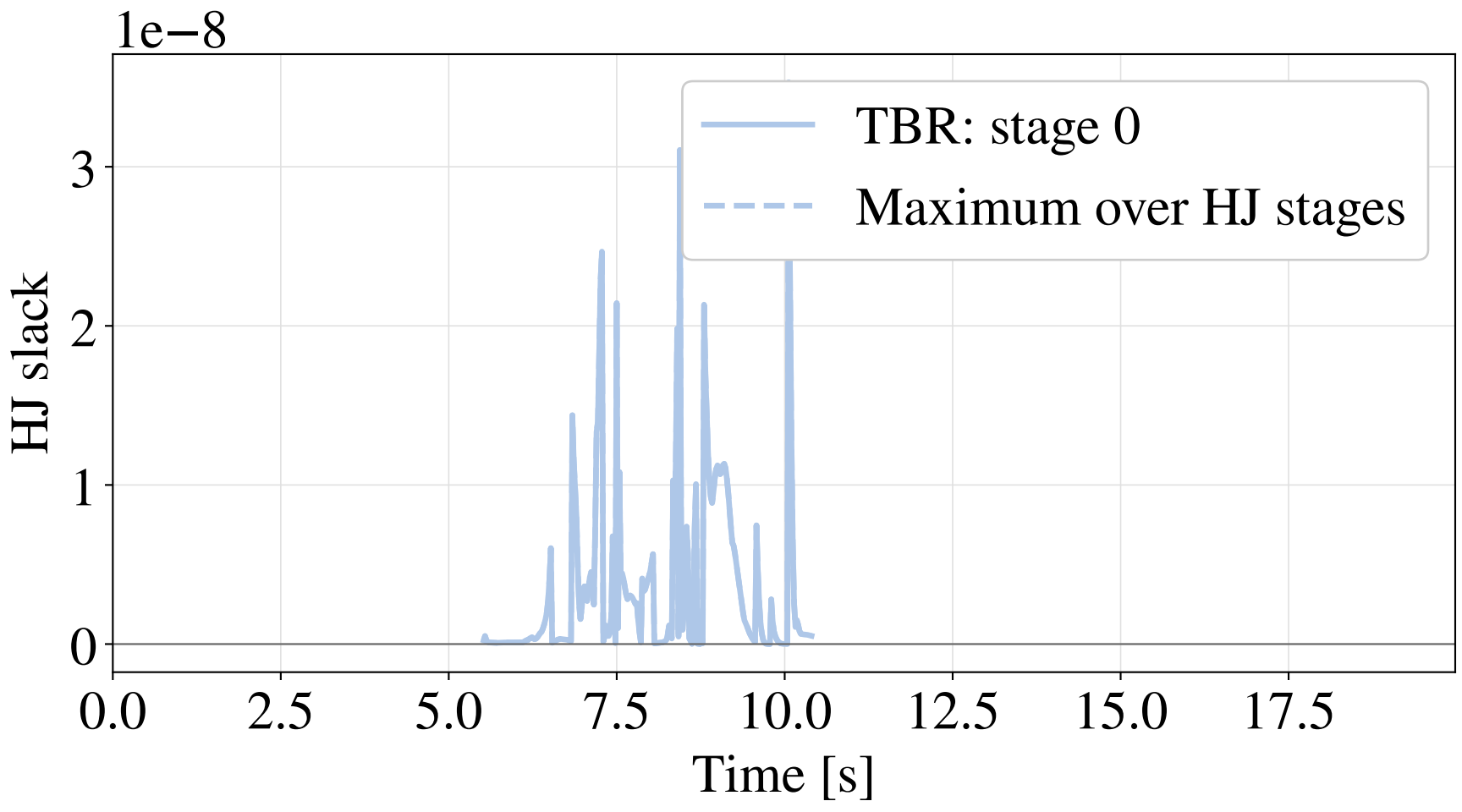

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ TBR
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
HJ TBR

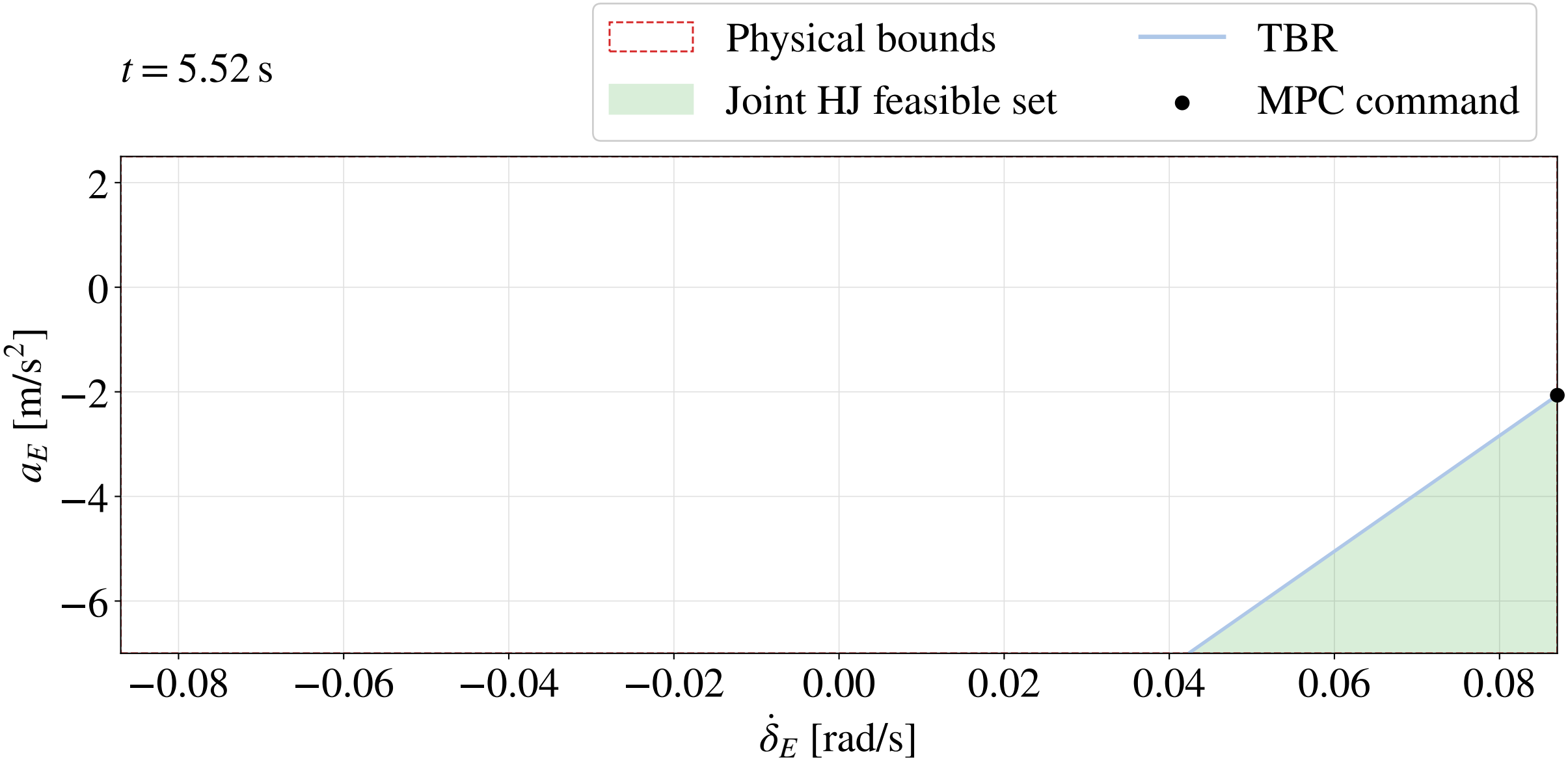

OCP cost
1.0
0.5
0.0
Time [s]
Total OCP cost
Cost components
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
Longitudinal
Velocity
Lateral
Collision
Overtake
Road left
Road right
Other
HJ slack

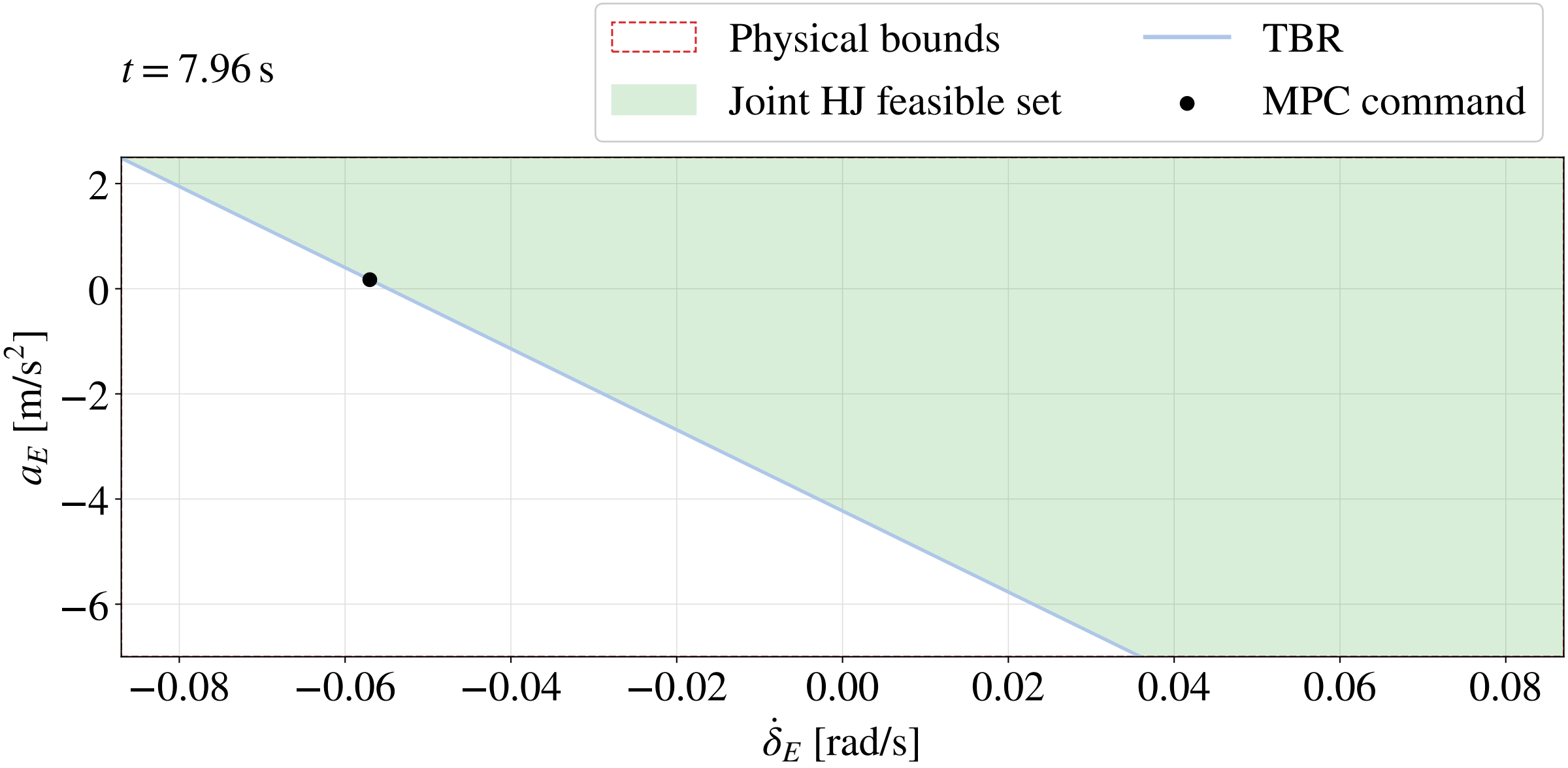

HJ residual
5
4
3
2
1
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
TTC: stage 0
Minimum over HJ stages

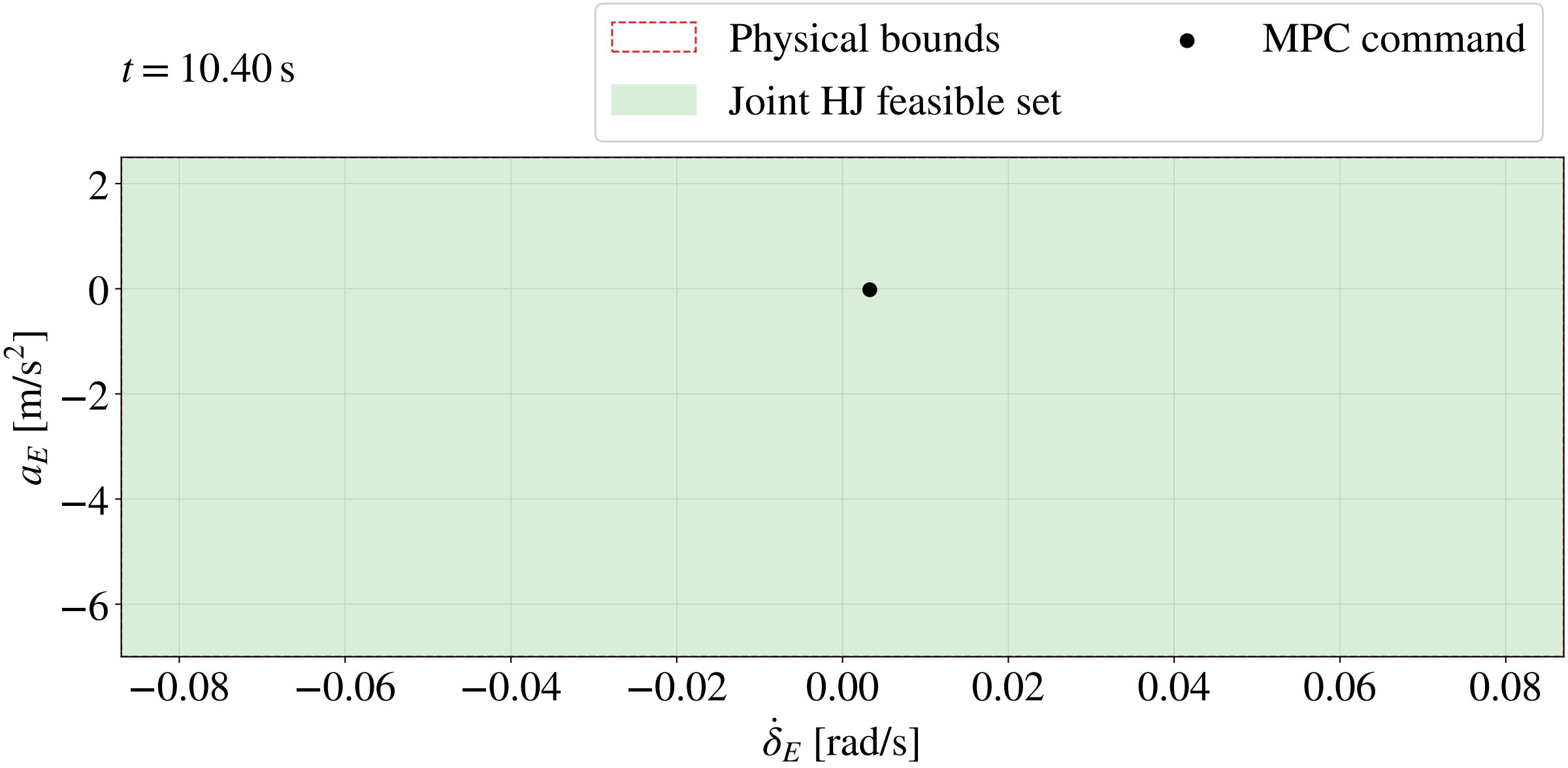

1e−8
3
2
1
0
HJ slack
TTC: stage 0
Maximum over HJ stages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]


HJ residual
0
1
2
3
4
5
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
TBR: stage 0
Minimum over HJ stages


1e−8
3
2
1
0
HJ slack
TBR: stage 0
Maximum over HJ stages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]


$t = 5.52$ s
Physical bounds
Joint HJ feasible set
TBR
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]


$t = 7.96$ s
Physical bounds
Joint HJ feasible set
TBR
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]


$t = 10.40$ s
Physical bounds
Joint HJ feasible set
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]